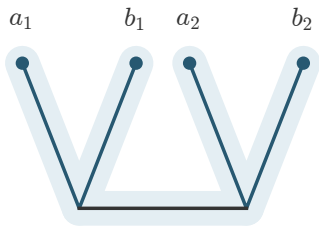
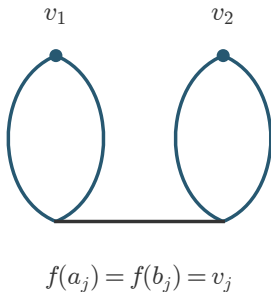


(a) Source tree Θ



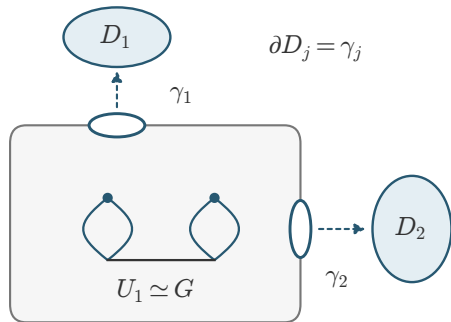
Thickening in L_0 :
a disk

(b) Quotient $G = f(\Theta)$



Paired endpoints
create two free cycles

(c) Exterior two-handle cores



Attach off the sheets:
 $U \simeq \text{pt}$, while $f^{-1}(U)$ is a disk